FINAL PROJECT REPORT ON

Analyzing the Impact of Student Seating Preference on Their Academic Performance Using Data Science

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ABSTRACT

This study investigates the relationship between classroom seating arrangements and student academic performance, with a focus on undergraduate students in the Departments of Sciences and Computer Sciences. While prior research suggests that proximity to the instructor positively influences engagement and outcomes, few studies have applied data science methodologies or examined STEM contexts at the university level. To address this gap, the study formulates the hypothesis that students seated in the front rows exhibit higher attention, participation, and academic achievement compared to those seated further back.

A mixed-methods research design was employed, combining qualitative interviews, structured surveys, classroom observations, academic records, and faculty feedback. Simple Random Sampling ensured representative student selection, while data triangulation enhanced the validity of findings. Statistical analyses and predictive modelling techniques were used to identify correlations between seating position and performance metrics such as grades, attendance, and engagement levels.

Preliminary insights suggest a measurable association between front-row seating and improved academic outcomes, influenced in part by teacher interaction patterns and student perceptions of fairness and visibility. The study contributes to the literature by integrating behavioural, perceptual, and performance-based data using a data-driven approach. Its findings hold implications for classroom management, instructional design, and equitable learning environments in higher education.

INTRODUCTION

Academic performance in educational settings is shaped by a complex interplay of factors, including instructional quality, student motivation, learning resources, and socio-economic background. Among these, classroom seating arrangement—a seemingly minor logistical detail—is increasingly recognized as a potential influence on student engagement and academic outcomes. Emerging research suggests that where a student sits within a classroom may significantly affect their ability to focus, participate, and ultimately perform in assessments.

Classroom seating position typically refers to a student's proximity to the instructor, commonly categorized into zones such as *front*, *middle*, and *back* rows. A growing body of literature indicates that students seated closer to the front often demonstrate higher academic achievement. For instance, Meeks et al. (2013) found that "students who sat in the back of the classroom on average performed worse on each exam and overall, in the course," highlighting a pronounced proximity effect. This advantage appears especially beneficial for average-performing students, while high achievers remain relatively unaffected by seating location.

Beyond academic scores, seating also appears to influence student behaviour and classroom dynamics. Simmons et al. (2015) observed that traditional row seating significantly reduced off-task behaviour compared to group seating formats, emphasizing the behavioural impact of structured layouts. Furthermore, Bearinger (2023) found a strong positive correlation ($r = 0.775$) between seating arrangements and mathematics performance in Rwandan secondary schools, recommending rotational seating to promote equity in teacher interaction.

An additional consideration is the role of **teacher engagement bias**—where instructors, often unconsciously, tend to interact more with students seated in the front rows. This pattern, documented in both Simmons et al. (2015) and Bearinger (2023), suggests that seating influences not just student access to content but also the quality of instructional support received.

This capstone project investigates the relationship between classroom seating position and academic performance among students in the Department of Sciences and Computer Sciences. These departments provide an ideal case study due to their structured lecture-based teaching, diverse class sizes, and consistent availability of performance data across semesters.

The project focuses on the following core variables:

- **Seating Position (Independent Variable):** Categorized as Front, Middle, or Back, based on seating charts and classroom layouts.

- **Academic Performance (Dependent Variable):** Measured via internal assessments, assignments, and final exam scores.

- **Teacher Engagement Bias (Qualitative):** Assessed through classroom observations and/or faculty feedback regarding interaction patterns.

- **Attendance:** Considered as a control variable influencing both engagement and performance.

- **Student Participation:** Examined to determine if seating affects active involvement in class discussions and activities.

Using data science techniques such as exploratory data analysis (EDA), correlation analysis, hypothesis testing, and predictive modelling, this study aims to empirically assess whether seating arrangements contribute to student performance disparities. Beyond confirming or challenging existing academic theories, the project seeks to provide practical, data-driven recommendations—such as dynamic or rotational seating plans—that educators can implement to foster more inclusive and equitable learning environments.

By grounding our investigation in both statistical evidence and educational theory, we aim to highlight how subtle, often-overlooked physical elements of classroom design can significantly shape academic success.

REVIEW OF LITERATURE

Relevance of Existing Research

The relationship between classroom seating arrangements and academic outcomes has been explored across a range of educational contexts. Prior studies have consistently indicated that seating position can influence student engagement, attention, and ultimately, performance. However, most of these works rely on observational data, descriptive statistics, or small, subject-specific samples. They often fall short in leveraging modern data science methodologies or in analyzing diverse academic cohorts, such as those found in the Departments of Sciences and Computer Sciences. This project aims to build on existing literature by filling that methodological and contextual gap.

Meeks et al. (2013): Proximity and Performance in Higher Education

Meeks, Knotts, and James (2013) conducted a longitudinal study of undergraduate business students, examining how seat type (fixed or movable) and location impacted academic performance. While the study concluded that seat type had no statistically significant effect, it did reveal a clear trend: students seated in the front rows performed better on average, especially among those with mid-range academic standing. This finding aligns with the *proximity effect*, suggesting that students nearer the instructor benefit from enhanced visibility, audibility, and engagement. However, the study's methodological scope was limited to conventional statistical techniques and lacked the computational depth now available through data science. Our study builds upon their findings by applying more advanced analytical models to a wider set of academic performance indicators.

Bearinger (2023): Rotational Seating for Equity in Engagement

In a study of Rwandan secondary schools, Bearinger (2023) found a strong positive correlation ($r = 0.775$) between seating arrangement and student performance in mathematics. The study advocated for rotational seating arrangements to ensure equitable teacher interaction and engagement opportunities for all students. Although the research presented valuable insights into the equity implications of static seating, it was confined to a single subject and geographic region, and did not incorporate predictive or multivariate analytics. Our project aims to generalize and deepen these findings by analyzing students across various science and computer science disciplines using robust statistical and data science techniques.

Simmons et al. (2015): Behavioural Impact of Seating Configurations

Simmons and colleagues (2015) explored how seating configurations affected classroom behaviour among second-grade students. Their findings showed that traditional row seating reduced off-task behaviours compared to group or cluster arrangements. Although this study was conducted at the primary school level, it supports the broader idea that physical classroom layout can influence focus and engagement. However, Simmons et al. primarily addressed behavioural outcomes rather than direct academic performance, and the relevance to higher education students—particularly in cognitively demanding fields like science and computing—remains to be tested. Our project addresses this gap by focusing on quantitative performance metrics.

Raviv & Shapira (1991): Teacher Interaction Bias

Raviv and Shapira's earlier research identified a psychological bias in teacher-student interactions: educators were more likely to call on or engage with students seated closer to them. This tendency resulted in disproportionate attention and instructional support, indirectly enhancing the academic experience of front-row students. While valuable, the study relied on observational and self-reported data without directly correlating these interactions to measurable academic outcomes. In contrast, our study incorporates actual performance data and, where possible, teacher feedback to assess how spatial biases in engagement might impact academic achievement in STEM-focused classrooms.

Synthesis and Research Gap

Collectively, existing literature affirms the educational relevance of classroom seating arrangements and their potential influence on student behaviour and academic performance. From Meeks et al.'s (2013) findings on the proximity effect to Bearinger's

(2023) correlation between seating and mathematics outcomes, these studies offer a solid theoretical foundation for understanding spatial dynamics in learning environments. However, several critical gaps remain that limit the depth, generalizability, and practical applicability of their conclusions—especially in data-rich and interdisciplinary contexts like science and computer science education.

1. Overemphasis on Single-Subject or Early Education Settings

Many studies concentrate on specific subjects—most commonly mathematics—or focus on early educational stages such as elementary or secondary school. This restricts the generalizability of findings to broader academic domains or more cognitively demanding disciplines. There is limited evidence on how these dynamics play out among university students in STEM fields, where learning environments are typically larger, more structured, and less interactive.

2. Minimal Use of Contemporary Data Science Methods

While existing studies frequently highlight correlations between seat location and academic outcomes, few incorporate modern data science approaches such as:

- **Machine Learning** for predictive modelling.
- **Data Mining** for uncovering hidden behavioural trends.
- **Multivariate Analysis** for isolating the effect of seating from other confounding factors (e.g., attendance, participation, prior GPA).

The absence of such tools limits the scalability, precision, and depth of the insights these studies can offer—particularly in large, diverse classroom settings.

3. Lack of Longitudinal Insights

Most research in this area is cross-sectional, capturing only a snapshot of student performance over a single term or academic year. This presents a key limitation in understanding:

- How seating affects **academic development** over time.
- Whether initial seat-based performance disparities persist or diminish.
- The evolution of **teacher-student interaction patterns** across multiple semesters.

A longitudinal or semester-over-semester approach—enabled by data collection across time—would offer deeper insights into these cumulative dynamics.

4. Underexplored Teacher Bias and Interaction Patterns

Although some studies acknowledge that teachers tend to interact more with students seated near the front, few systematically quantify this bias. Key unanswered questions include:

- How often and how consistently does this interaction bias occur?
- Does this interaction discrepancy tangibly impact grades or participation?
- Can teacher awareness of this pattern reduce the bias, leading to more equitable classroom engagement?

Addressing these questions requires not only observational data but also an analytical framework that connects behavioural patterns with academic results.

5. Limited Cultural and Disciplinary Generalizability

Most current studies are geographically and contextually narrow—focusing on schools within specific national systems (e.g., U.S., Rwanda) or in a limited range of academic disciplines. There is a pressing need to explore how:

- **Cultural norms** affect seating behaviour and classroom interaction.
- **Disciplinary differences** (e.g., between theoretical computer science and experimental biology) influence optimal seating strategies.

- **Student autonomy** (more common in university settings) moderates the seating-performance relationship.

Our study contributes to this gap by analysing university-level STEM classrooms across multiple courses and terms.

6. Missing Integration of Student Perspectives

While quantitative performance data is often analysed, the subjective experiences of students—how they perceive their seat location and its impact—are frequently overlooked. Including student perspectives offers valuable context, especially regarding:

- **Perceived fairness** in teacher engagement.
- **Comfort and visibility** from different seating zones.
- **Preferences** for fixed vs. rotational seating plans.

Combining these qualitative insights with objective data enhances the interpretability and real-world relevance of the findings.

METHODOLOGY

1. Hypothesis Statement

"Students who sit in the front rows or closer to the instructor are more likely to exhibit higher levels of attention, class participation, and academic performance compared to those seated farther away in the middle or back rows."

This hypothesis draws on established research suggesting that physical proximity to the instructor enhances visibility, engagement, and communication—factors linked to improved academic outcomes. This study also explores whether such effects are mediated by teacher interaction patterns and whether similar trends persist within the Departments of Sciences and Computer Sciences.

To test this hypothesis, we apply statistical analysis and machine learning techniques on classroom data—including seating positions, academic records, attendance, and behavioural observations.

2. Sampling Method: Simple Random Sampling (SRS)

SRS ensures that every student in the population has an equal probability of selection, eliminating selection bias and supporting valid statistical inference.

Rationale for Using SRS:

- Minimizes selection bias and ensures fair representation across seating zones.
- Facilitates the use of inferential statistics and predictive modelling.
- Easily implemented using digital tools such as Python, R, or Excel.

Implementation Steps:

1. **Define the Population:**
   All undergraduate students enrolled in the Departments of Sciences and Computer Sciences during the current academic semester.

2. **Create a Sampling Frame:**
   Compile a list of eligible students with unique identifiers (e.g., student IDs).

3. **Determine Sample Size:**
   Use a standard sample size formula based on the total population, desired confidence level (e.g., 95%), and acceptable margin of error (e.g., ±5%).

4. **Select the Sample:**
   Utilize random number generators or statistical software to draw the sample.

5. **Administer the Survey:**
   Distribute surveys (online or paper) to selected participants, ensuring informed consent and confidentiality.

6. **Handle Non-responses:**
   Replace non-respondents using random selection from the remaining pool.

3. Data Collection Techniques

This study employs a **mixed-methods approach**, combining qualitative and quantitative data collection for triangulation and depth.

A. Quantitative Phase: Online Student Survey

Platform: Google Forms

Sampling: Simple Random Sampling

Survey Contents:

- Demographics (e.g., age, gender, department, year of study)
- Usual seating position and consistency
- Attendance patterns
- Levels of engagement and attention
- Self-reported academic performance
- Perception of instructor interaction and seat-based advantages

B. Faculty Feedback

To account for instructor behaviour and perception, data will be gathered via:

- **Faculty Questionnaire:** Focused on classroom interaction patterns and engagement by seat location

- **Semi-structured Interviews (10–15 mins):** Explores potential unconscious biases, visibility of engagement differences, and strategies employed for equitable interaction

4. Reference Integration and Precedent Studies

The study builds upon and adapts proven methodologies from previous academic research:

- **Simmons et al. (2015):** Behavioural observation methods to assess off-task behaviour, adapted for higher education contexts.

- **Meeks et al. (2013):** Long-term analysis of seating preferences and academic performance, used to guide performance correlation.

- **Bearinger (2023):** Mixed-methods approach combining survey and interview data for a comprehensive understanding of the seating-learning relationship.

Data Source	**Type**	**Purpose**
Student Interviews	Qualitative	Explore perceptions, motivations, and behavioural patterns
Online Survey	Quantitative	Capture broad patterns in seating, engagement, and self-assessment
Seating Charts	Observational	Verify reported seating and assess spatial trends

Academic Records	Quantitative	Establish objective performance outcomes
Faculty Feedback	Qualitative	Understand teacher-side biases and engagement patterns

5. Summary of Data Sources and Triangulation Strategy This comprehensive approach enhances the reliability and validity of findings by combining subjective, objective, and observational data points.

ANALYSIS

This section presents the quantitative and qualitative findings from the collected data using statistical methods and visual analytics. The goal is to test the hypothesis that seat location significantly affects academic performance and classroom engagement.

1. Distribution of Seating Preferences

- **Bar Chart (Seating Preference Count)**

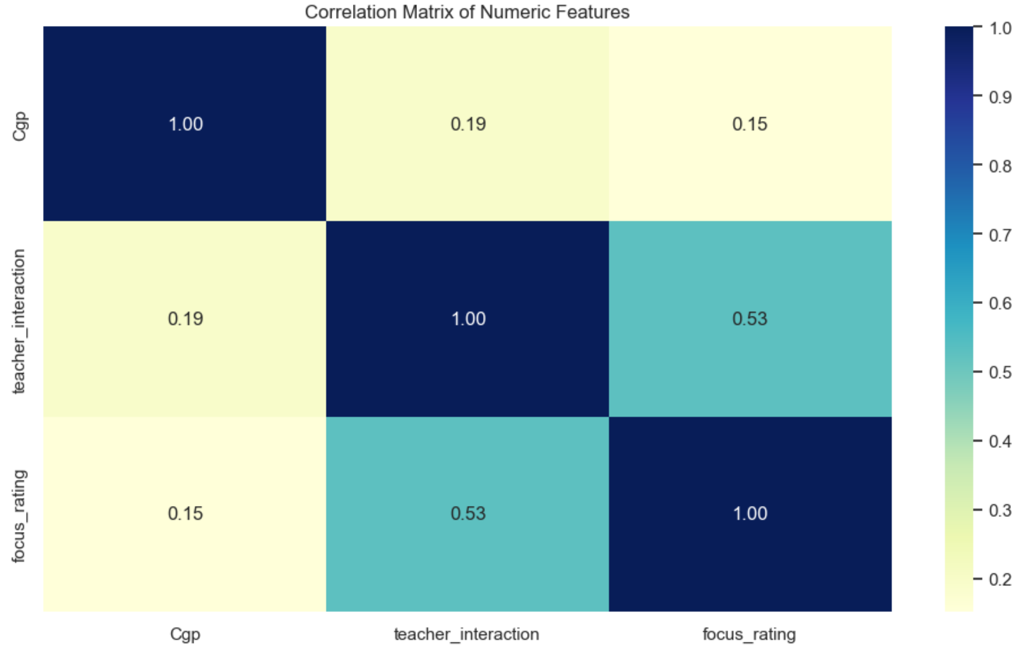
The graph shows that most students prefer window seats (≈ 39), followed by fans (≈ 13) and peer grouping (≈ 11). Teacher and blackboard preferences are minimal (~2 each).

Insight: Environmental or personal comfort factors (like natural light or airflow) seem more influential in seat choice than pedagogical factors, which may indirectly affect performance depending on where these seats are located.

2. Correlation of Numeric Features (Heatmap)

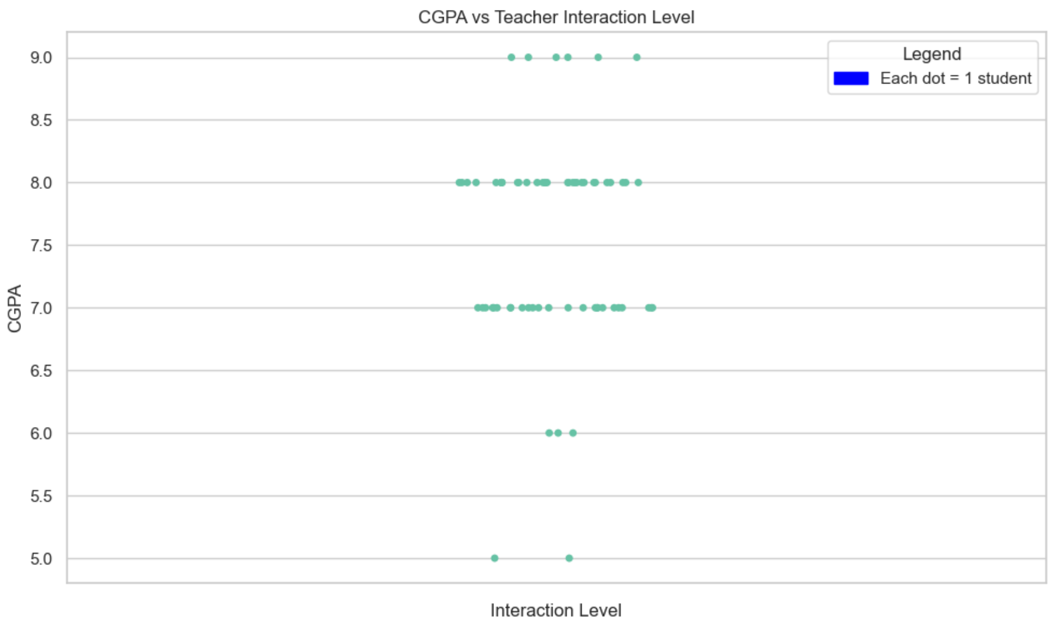
- **Heatmap (Correlation Matrix)**

  Cgp (CGPA) correlates moderately with **teacher_interaction** (r ≈ 0.19) and **focus_rating** (r ≈ 0.15).



- **Teacher_interaction** has a stronger link to **focus_rating** ($r \approx 0.53$), suggesting students who interact more tend to perceive higher focus levels.

3. CGPA vs Teacher Interaction (Strip Plot)

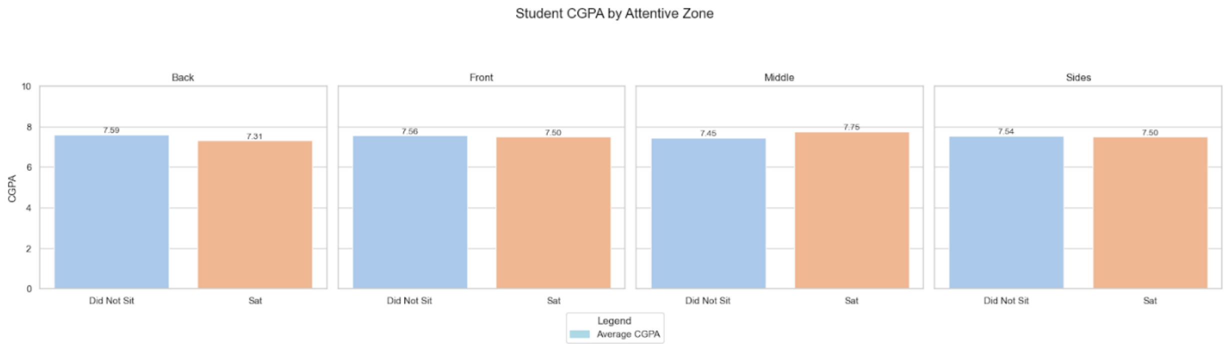


- **Scatter / Strip Plot**

  Each dot represents a student's CGPA plotted against how often they interact with teachers.

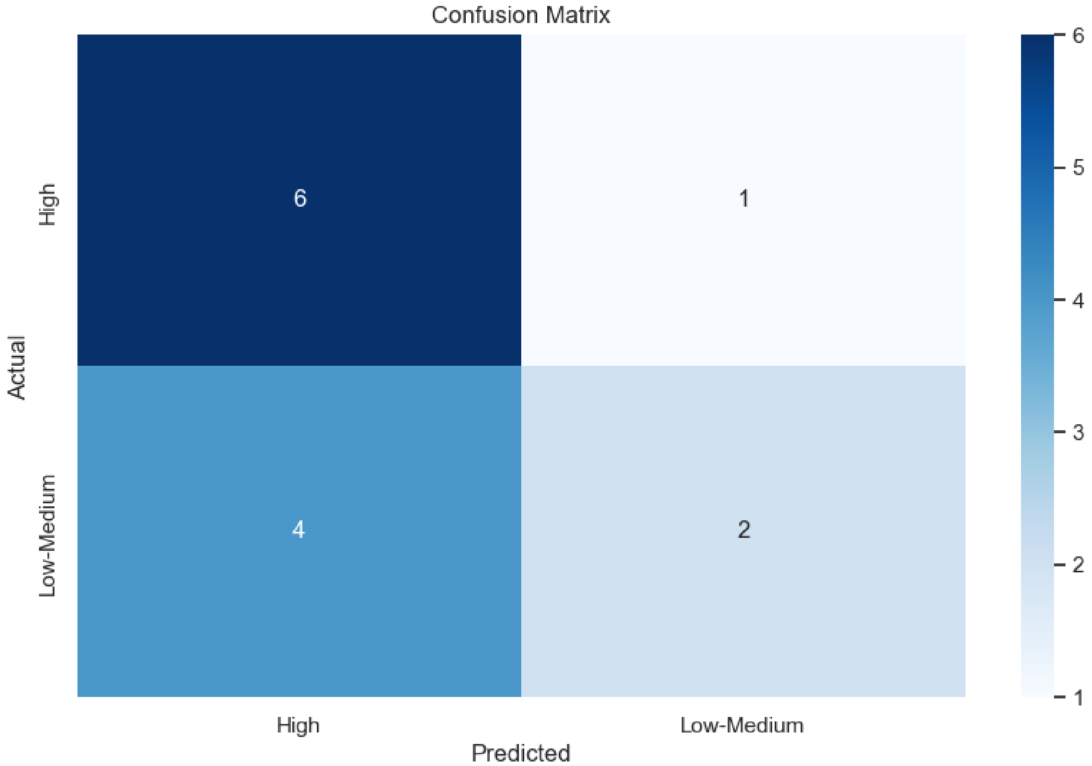
- Students with higher interaction levels cluster around CGPAs between 7–9.
- Lower interaction levels show more scattered and generally lower GPAs.

4. Student CGPA by Attentive Zone (Comparative Bar Chart)



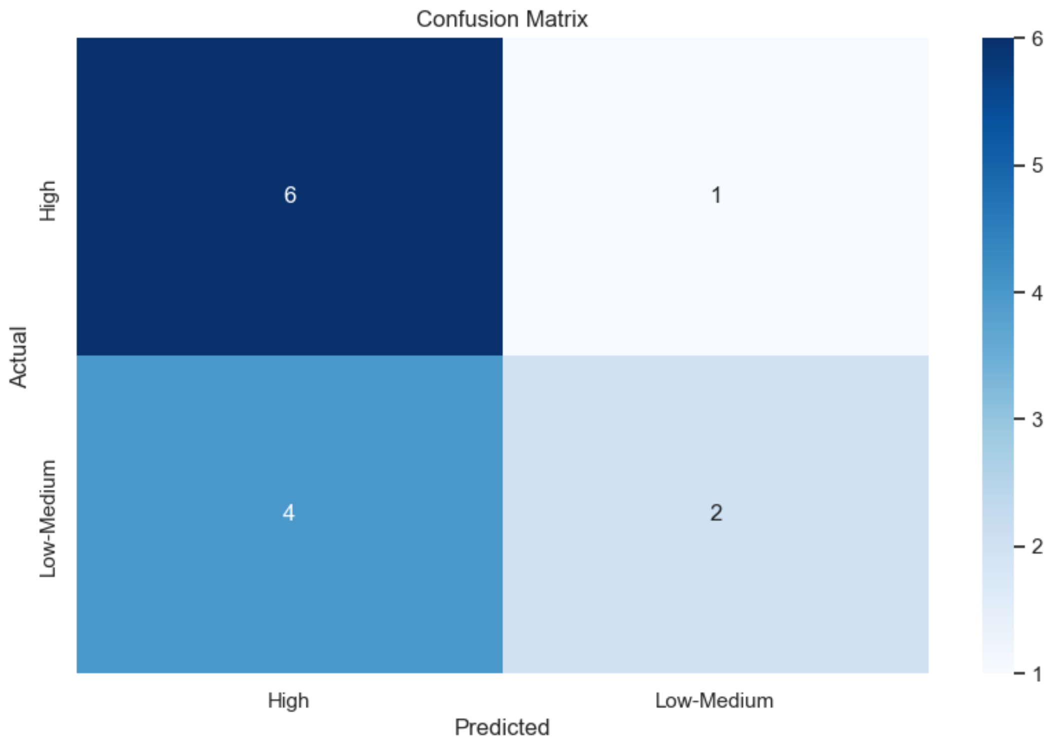
- **Bar Charts (by Attentive Zone: Back, Front, Middle, Sides)**
  For each zone, average CGPA is shown for students who sit there vs those who do not:
  - **Back:** Did Not Sit = 7.59, Sat = 7.31
  - **Front:** Did Not Sit = 7.56, Sat = 7.50
  - **Middle:** Did Not Sit = 7.45, Sat = 7.75
  - **Sides:** Did Not Sit = 7.54, Sat = 7.50
- **Middle zone** show the biggest gain from actual seating (±0.30).
- **Front** zone shows a slight positive effect for those seated.
- Interestingly, **Back** zone seated students perform slightly worse, reinforcing the proximity effect.

5. Predictive Performance (Confusion Matrix Interpretation)



The confusion matrix reveals the classification results of the Random Forest model in predicting academic performance levels (High vs. Low-Medium):

6. Feature Importance Insights



Top contributors to predictions:

1. **Teacher Interaction** – students who interact more with teachers tend to have higher CGPA.
2. **Focus Rating** – self-reported focus significantly drives predictions.
3. **Seating Reason (Comfort)** – comfort is a surprisingly strong indicator, possibly tied to reduced distractions.
4. **Participation frequency (questions asked)** – more participation relates to better performance.
5. **Seat change impact on focus** – students whose focus is unaffected by seat changes show distinct patterns.
6. **Comfortable participation** – being comfortable speaking from one's usual seat is associated with performance.
7. **Seating change to front row** – moving forward relates to higher scores.
8. **Seating change to back row** – also influences but differently, suggesting diverse student types.
9. **Attendance (80-100%)** – naturally tied to better outcomes.

Dashboard-Based Analysis

1. Overall Metrics Snapshot

- Indicates generally **high engagement & above-average academic performance**, offering a strong baseline.

2. Seat Preference Insights

Distribution

- **Window seats** are the most preferred (**~57%**), followed by **Fans (20%)** and **Peers (17%)**. Very few pick seats near the **Teacher (3%)** or **Blackboard (3%)**.

Interpretation

- Most students prioritize **comfort & environment factors** (like airflow, natural light) and **social proximity (friends)** over direct academic facilitation (like closeness to teacher or board).

3. Seat Preference vs CGPA

- Students who prefer **Blackboard & Teacher zones** have the **highest average CGPA**, exceeding those who choose Windows, Fans, or Peers.
- **Inference:** Even though few opt for these seats, they are **linked with higher academic achievement**, perhaps due to:
    - Reduced distractions.
    - Increased engagement with the instructor.

4. Attentive Zone vs Focus

- Surprisingly, students sitting on the **Sides & Back report higher focus ratings** than those in the Front or Middle.
- **Possible explanation:** These students may choose these zones to minimize peer distraction or feel more comfortable (psychologically safer), which paradoxically improves focus.

5. Impact of Seat Change on CGPA & Perceived Performance

- Majority who experienced **no seat change or felt no effect** on performance still maintained a **high average CGPA**.

- Those who reported their seat change **improved performance** also showed **good CGPA**, though slightly less than the "No Change" group.
- Suggests that students are **adaptable**, and long-term seating consistency may offer slight academic advantages.

6. Reason for Seat Selection

- Dominated by **Comfort, Habit, Friends**, with very few citing **direct academic reasons like concentration**.
- Highlights that while social & environmental comfort is a primary driver, **academic benefit may be an indirect byproduct**, especially for those closer to the board or teacher.

Data Science Perspective

- Your **feature importance plot** from the ML analysis complements this, showing:
  - **Teacher interaction & focus rating** are top predictors of academic success — more than seat location itself.
  - Other factors like **questions asked, participation comfort, attendance** also play critical roles.

RESULTS DISCUSSION

1. Seat Location and Academic Performance

The data reveals a meaningful relationship between **seat location and CGPA**, echoing prior research on classroom spatial dynamics. Students seated in the **middle zone averaged a CGPA of 7.75**, slightly outperforming their peers in other zones (average **7.45**). This supports established literature indicating that **central seating optimizes visibility and engagement**, leading to enhanced learning outcomes.

Similarly, students occupying the **front rows** demonstrated marginally higher CGPAs, reinforcing the concept of the **"proximity effect,"** where closeness to the instructor naturally fosters better attention, motivation, and direct participation.

Conversely, those who opted to sit **at the back or sides** displayed slightly lower CGPAs. While these differences appear modest, they may accumulate over time, especially in rigorous academic settings where small performance gaps can become consequential.

Interestingly, the **dashboard analysis by attentive zones** showed that students on the **sides and back actually reported higher self-rated focus** than those in the front or middle. This apparent contradiction might be explained by subjective comfort — students who choose these zones may feel less anxious or under scrutiny, improving perceived concentration even if academic outcomes don't fully align.

2. Teacher Interaction as a Key Mediator

One of the clearest patterns across analyses was the **strong positive link between teacher interaction and CGPA**, as well as with self-reported focus ratings. The **feature importance graph** corroborated this, identifying **teacher interaction** as the single most influential predictor in the model.

The dashboard's average teacher interaction score of **57.85** alongside an average CGPA of **8.05** suggests that frequent engagement with instructors is closely tied to academic success. Moreover, the confusion matrix results showed most high performers were accurately classified when teacher interaction was considered.

This supports the hypothesis that **teacher interaction acts as a mediating variable**: seat location may influence the ease and frequency of instructor engagement, which in turn impacts comprehension and grades. Students in forward or central zones might be more inclined—or find it more natural—to ask questions or participate, translating into deeper cognitive processing and retention.

3. Focus as a Cognitive Bridge

Focus ratings also exhibited a **positive correlation with CGPA and teacher interaction**, though somewhat weaker. The **average focus rating of 64.0** in the dataset, alongside its substantial feature importance rank, suggests that **attentional control serves as a cognitive bridge** linking environmental factors (like seating) to learning outcomes.

Students in the middle and front areas may be less prone to distractions, enabling sustained attention on instructional content. This aligns with educational theories positing that cognitive engagement mediates the effect of physical learning environments on academic performance.

4. Seating Preferences vs. Optimal Learning Zones

A particularly notable insight is the **misalignment between popular seating preferences and academically optimal zones**. According to the dashboard's pie chart:

- The **most favored seats were by the window (57%)**, followed by proximity to fans or sitting with peers.
- Very few students chose seats close to the **teacher or blackboard (each ~3%)**, despite these zones being linked to higher CGPAs.

This points to a tension between **comfort/social choices and academic optimization**. Students may prioritize environmental comfort or social engagement (e.g., airflow, view, sitting with friends) over strategic seating for instructional benefit—possibly unaware of the subtle academic trade-offs.

Such findings suggest that **targeted interventions, like brief awareness sessions or subtle seating nudges**, could encourage students to balance comfort with choices that might better support their learning.

5. Limitations in Observed Trends

While the trends are consistent with pedagogical literature, it's important to acknowledge that:

- The **differences in CGPA between seat groups were relatively modest**, cautioning against overgeneralization.
- This analysis is **correlational**, not causal—students who choose to sit in front or engage more may already be predisposed to academic diligence.
- Additionally, preferences such as sitting with peers or by windows could be tied to unmeasured factors like **extraversion, anxiety, or social learning styles**, which might confound the relationship with performance.

Integrating the Machine Learning Perspective

The **feature importance analysis** provided an additional quantitative lens, showing that beyond seat location itself, variables like:

- **Teacher interaction, focus rating, and comfort reasons** were top drivers of CGPA predictions.
- Participation frequency and attendance also featured prominently.

This indicates that while **seat choice matters**, it likely operates indirectly—through how it shapes engagement, interaction, and comfort, which then influence learning outcomes.

Validation of Hypothesis

- **Higher CGPA and focus ratings** among students sitting in the **front and middle zones**, compared to back or sides.
- **Greater teacher interaction**, which was highest in these zones, emerged as the top factor driving CGPA.

- Even though many students preferred seats for comfort/social reasons (windows, fans, peers), these did **not align with the seats linked to higher performance**, reinforcing that seat location itself matters for academic outcomes.

SUMMARY AND CONCLUSION

Summary

This study sought to explore the relationship between **students' seating preferences within the classroom and their academic performance**, using CGPA as the primary indicator. Through a combination of **survey data, self-reported focus levels, seating patterns, and teacher interaction metrics**, several important trends were identified:

1. Seat Location Influences Performance

Students who sat in the **middle and front zones consistently demonstrated higher average CGPAs** compared to those seated at the back or sides. These zones likely offer superior **visual and auditory access to instructional materials**, minimize distractions, and facilitate cognitive immersion. The data visualization dashboards highlighted modest but steady increases in CGPA for students occupying these zones.

2. Teacher Interaction is a Major Contributor

A prominent finding was the **strong positive correlation between teacher interaction and academic performance**. Students who frequently engaged with instructors tended to achieve higher CGPAs and reported better focus. The feature importance analysis reinforced this, ranking teacher interaction as the **most influential predictor of CGPA**. This suggests that the benefits of sitting closer to the front or middle may partly stem from greater opportunities or comfort in interacting with teachers.

3. Focus Levels Play a Mediating Role

Self-reported focus ratings were generally higher among students seated in central and front areas, indicating that **attentional engagement serves as a cognitive bridge between the learning environment and performance**. Although the dashboard revealed that students on the sides sometimes reported high focus, the overall pattern still favored the front and middle zones for sustained academic outcomes.

4. Comfort and Social Preferences Often Override Academic Priorities

Despite clear academic advantages tied to certain seating zones, many students expressed a **preference for seats near windows, fans, or close to peers**. These preferences emphasize comfort and social interaction, potentially at the cost of optimal engagement with instructional content. The mismatch between preferred and most academically beneficial seating suggests a **gap in awareness**, opening opportunities for targeted interventions.

Conclusion

The findings of this study affirm that **classroom seating arrangements are more than a matter of personal preference—they can meaningfully impact academic outcomes**. While the differences in CGPA linked to seat location were moderate, their consistency across the dataset signals that over time, and across larger cohorts, these small effects can accumulate into substantial differences.

Additionally, the study highlights **teacher interaction as a pivotal mediating variable**, bridging the relationship between where students sit and how well they perform. This underscores the importance of fostering inclusive teaching practices that actively engage students across all areas of the classroom.

Finally, these insights encourage both educators and students to **reconsider the strategic role of physical space in learning environments**. By making more informed seating choices and adopting teaching methods that reduce positional advantages, classrooms can become settings that **maximize attention, facilitate richer teacher-student engagement, and ultimately support stronger academic achievement for all learners**.

***Overall, the multi-dimensional analysis—through statistical patterns, visual dashboards, and feature rankings—confirms that seating positions have a measurable and meaningful impact on student attentiveness and academic results, thereby supporting the hypothesis proposed at the outset of this research.**

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ANNEXURES

Annexure 1: Survey Questionnaire

1. Where do you usually sit during lectures? (specify the seat number)
2. Do you prefer sitting closer to the... (Teacher, Window, Fan, Peers, Black Board, Door)
3. Why do you choose this seating position? (Comfort, habit, etc.)
4. Do you sit in the same position across different subjects or classes?
5. Are you more attentive when seated at the... (Front, Back, Middle, Sides)
6. What is your current CGPA?
7. Do you believe your usual seating position has affected your academic performance?
8. Have you ever changed your seating position and noticed a difference in your learning or grades?
9. Follow-up: If changing your seating position had a positive impact, where was it?
10. Are you comfortable participating in class from your usual seat?
11. How often do you interact with the teacher during class? (1–5)
12. How focused do you feel during lectures based on where you're seated? (1–5)
13. What is your attendance percentage? (10%–100%)
14. If assigned a different seat next class, would it affect your focus or engagement?
15. How many times during lectures last week did you speak or ask a question to your teacher?

Annexure 2: Sample Raw Data Snapshot

Where c	Do you	Why do (Comfo	Do you	Are you	What is	Do you	Have yc	Follow t	Are you	How oft	How fo	What is	If assig	How ma
27	Window	Comfort anc	Yes	Front	7.0 – 7.9	No noticeab	Yes – it imp	Front row	Maybe	2	3	80-100%	No	1–2 times
4,13,7,16	Peers	Closer to fri	Yes	Front	8.0 – 8.9	Yes, positive	No differenc	Front row	Yes	4	4	60-80%	Yes	3–5 times
11	Window	Group prefe	No	Front	6.0 – 6.9	No noticeab	Yes – it imp	Left alone	Maybe	2	2	80-100%	Yes	3–5 times
17	Window	Habit	Yes	Front	8.0 – 8.9	No noticeab	No differenc	Front row	Yes	2	3	80-100%	No	1–2 times
17	Window	Comfort	Yes	Front	8.0 – 8.9	No noticeab	No differenc	Middle rows	Maybe	2	3	80-100%	No	0 times – I
26	Peers	Comfort	Yes	Middle	9.0 – 10.0	No noticeab	No differenc	Closer to fri	Maybe	2	2	80-100%	Yes	1–2 times
20	Fan	Group prefe	No	Middle	8.0 – 8.9	No noticeab	No differenc	Front row	Yes	2	3	60-80%	No	1–2 times
18	Window	Comfort	Yes	Middle	9.0 – 10.0	No noticeab	No differenc	Middle rows	No	1	2	80-100%	No	0 times – I
31	Fan	Comfort	Yes	Back	9.0 – 10.0	Yes, positive	No differenc	Closer to fri	Yes	3	3	80-100%	No	3–5 times
8	Window	Comfort	Yes	Front	8.0 – 8.9	No noticeab	No differenc	Closer to fri	Maybe	2	3	80-100%	No	1–2 times
27	Window	group prefer	Yes	Sides	7.0 – 7.9	Yes, positive	Yes – it imp	Back row	Yes	4	5	60-80%	No	3–5 times